

iter18 -> iter19 · July-05 - 1B replication + scale transfer

Surgery >> frozen HOLDS on the 1B (ViT-g) AND replicates 2B -> 1B

=> greenlights the iter19 full-scale (116k) run on the cheaper 1B backbone · n=1,825 · 95% BCa CI

0 · Data sources (Task 1) - every figure -> its eval_metrics.{csv,json}

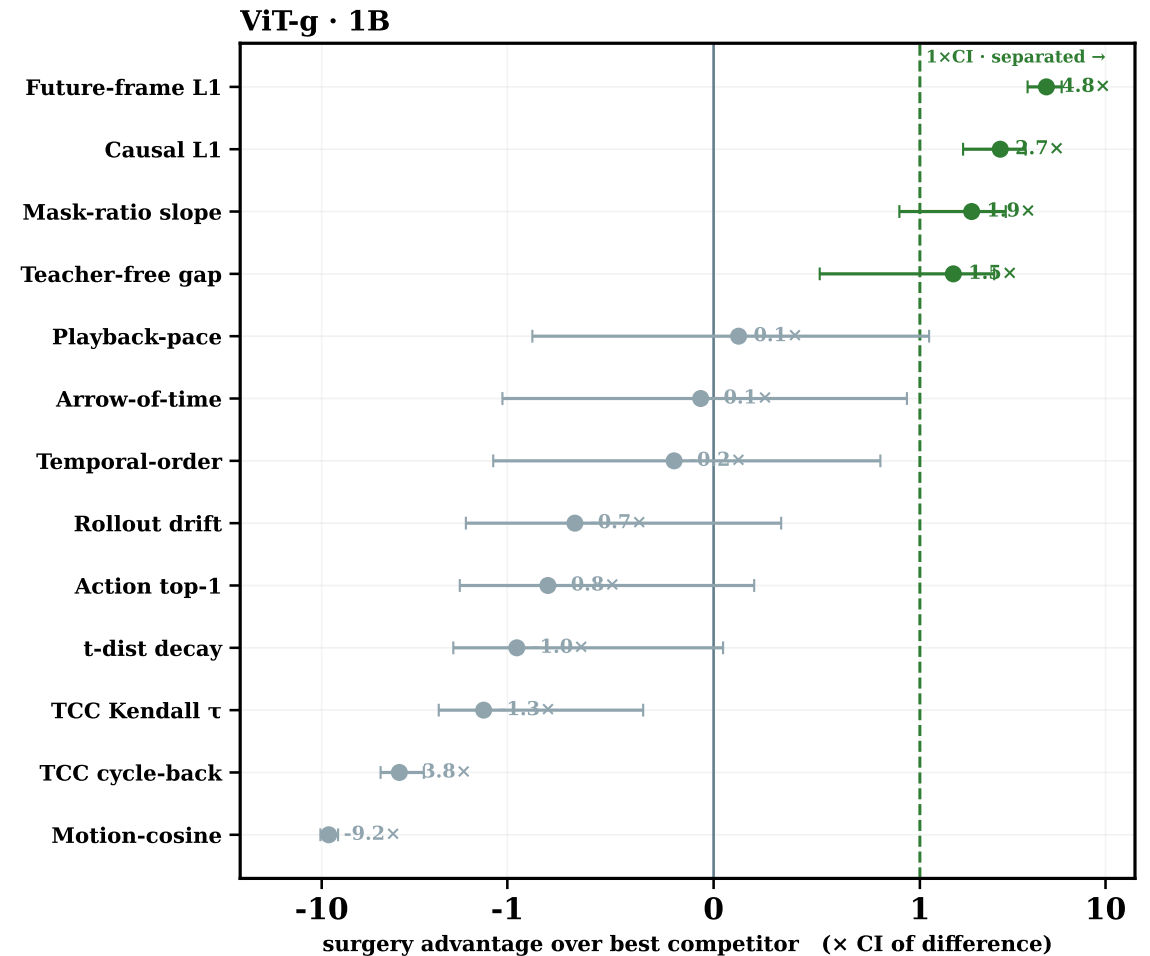
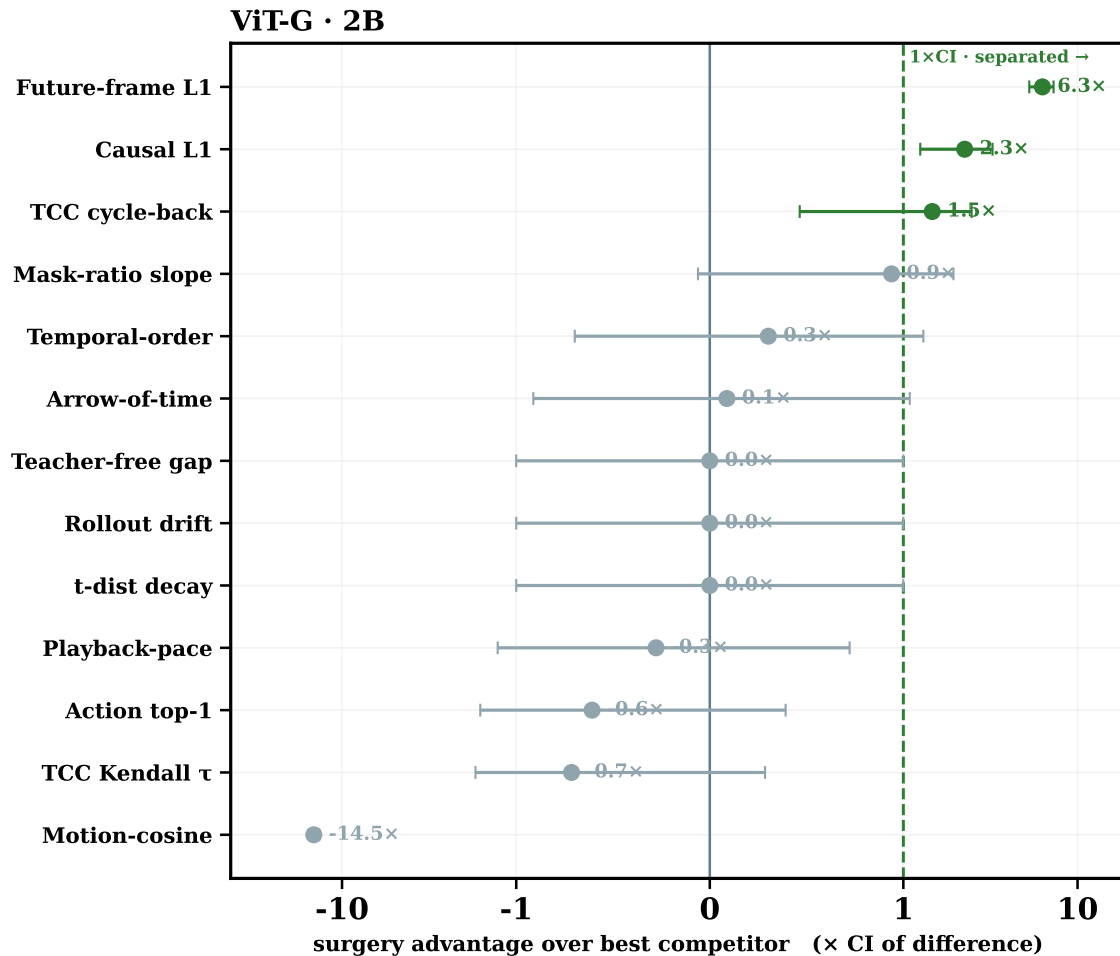
figure	what it compares	source (m13 --cross-plots)
forest_plot_best_ci / _mean	OURS vs BEST competitor (stat + magnitude)	both eval_metrics.json (plot_forest, vs=best)
forest_plot_frozen_ci	OURS vs FROZEN (the paper claim)	both eval_metrics.json (plot_forest, vs=frozen)
scale_replication	1B rank vs 2B rank (Spearman rho)	both eval_metrics.json (plot_scale_replication)
eval_scorecard_combined	2B champion stacked over 1B	the two per-backbone eval_scorecard.pdf

canonical per-backbone metrics (under iter18_.../result_outputs/):

- 2B v5_1B/poc/vjepa_2_1_vitG_2B/eval/eval_10k/probe_plot/metrics_watch/vjepa_2_1_vitG/eval_metrics.{csv,json}
- 1B v5_1B/poc/vjepa_2_1_vitg_1B/eval/eval_10k/probe_plot/metrics_watch/vjepa_2_1_vitg/eval_metrics.{csv,json}

1 · Surgery vs the STRONGEST competitor - statistical (CI-widths)

Forest plot — does surgery statistically separate from the best competitor? (green past dashed line = yes · surgery_raw ∈ OURs)

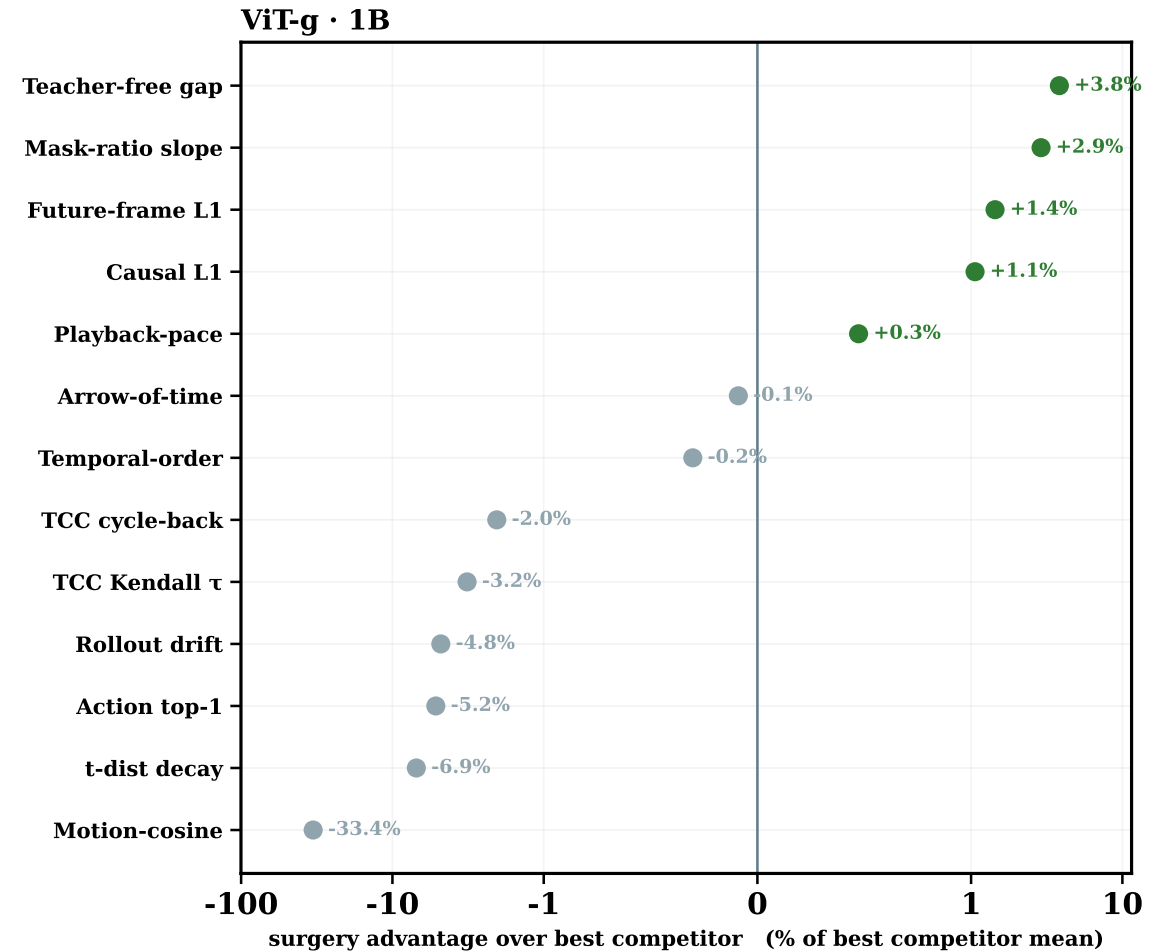
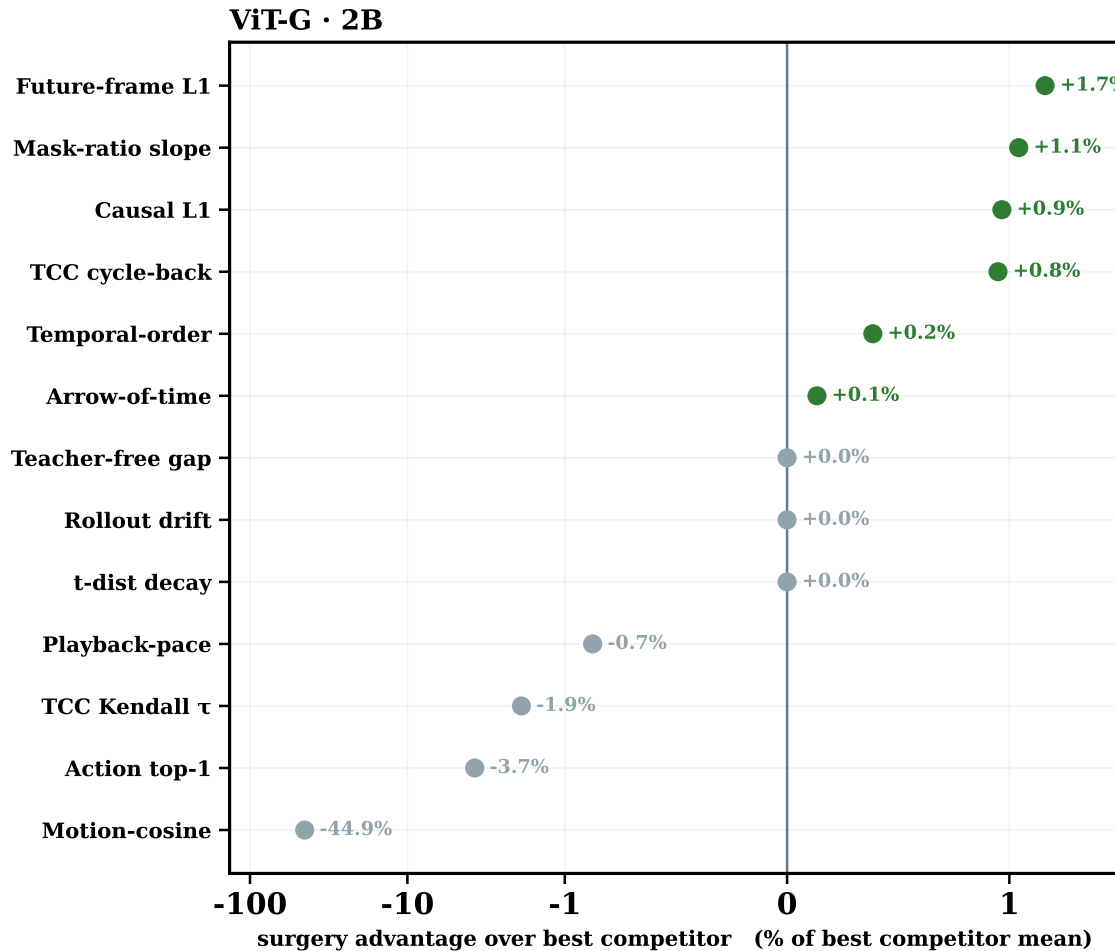


● Hardest bar (vs the best NON-ours arm, not frozen): future-L1 6.3x/4.8x and causal-L1 2.3x/2.7x CI-widths clear significance at 2B/1B.

● Motion-cosine sits far left (-14.5x / -9.2x) -- surgery cedes semantic separation to full-FT; reported, not hidden.

2 · Surgery vs the strongest competitor - effect size (raw %)

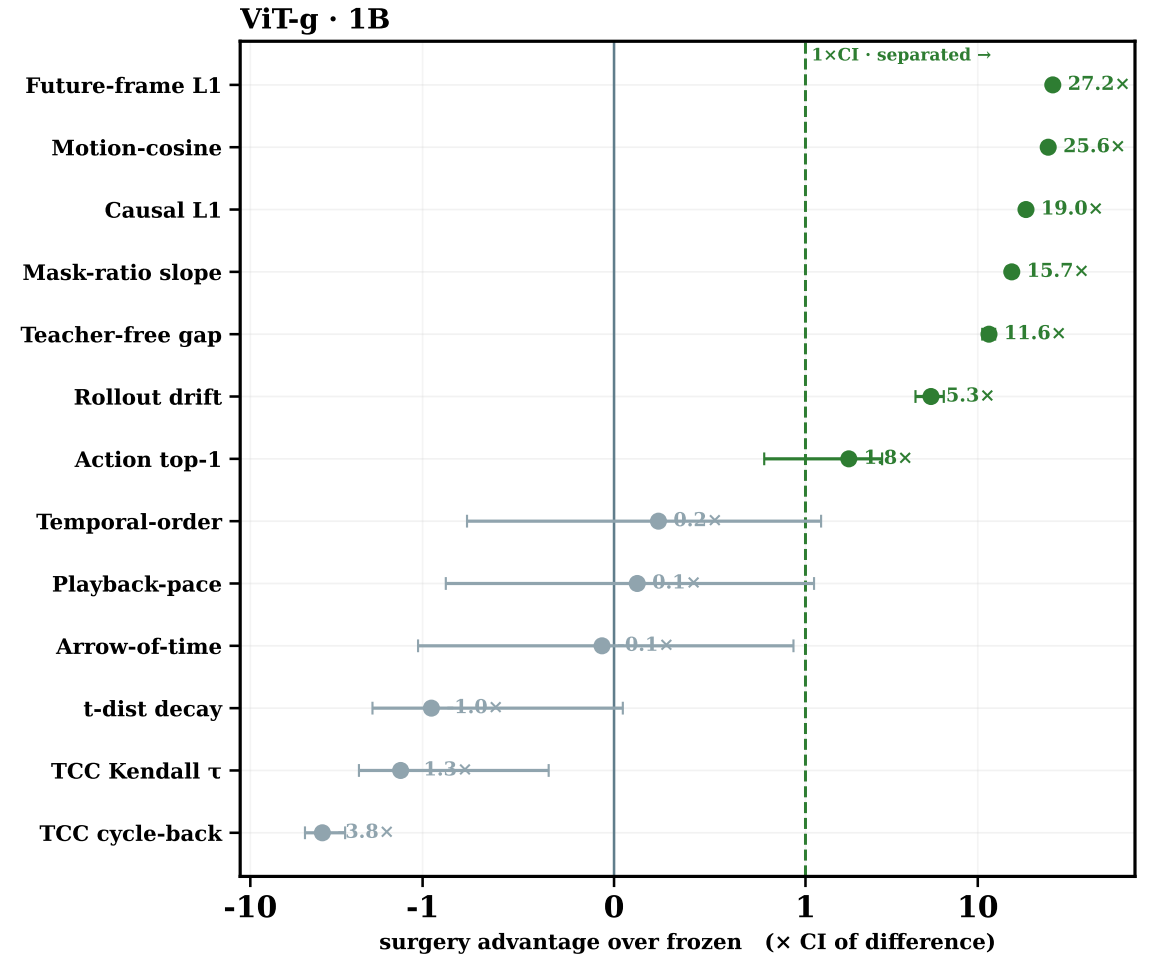
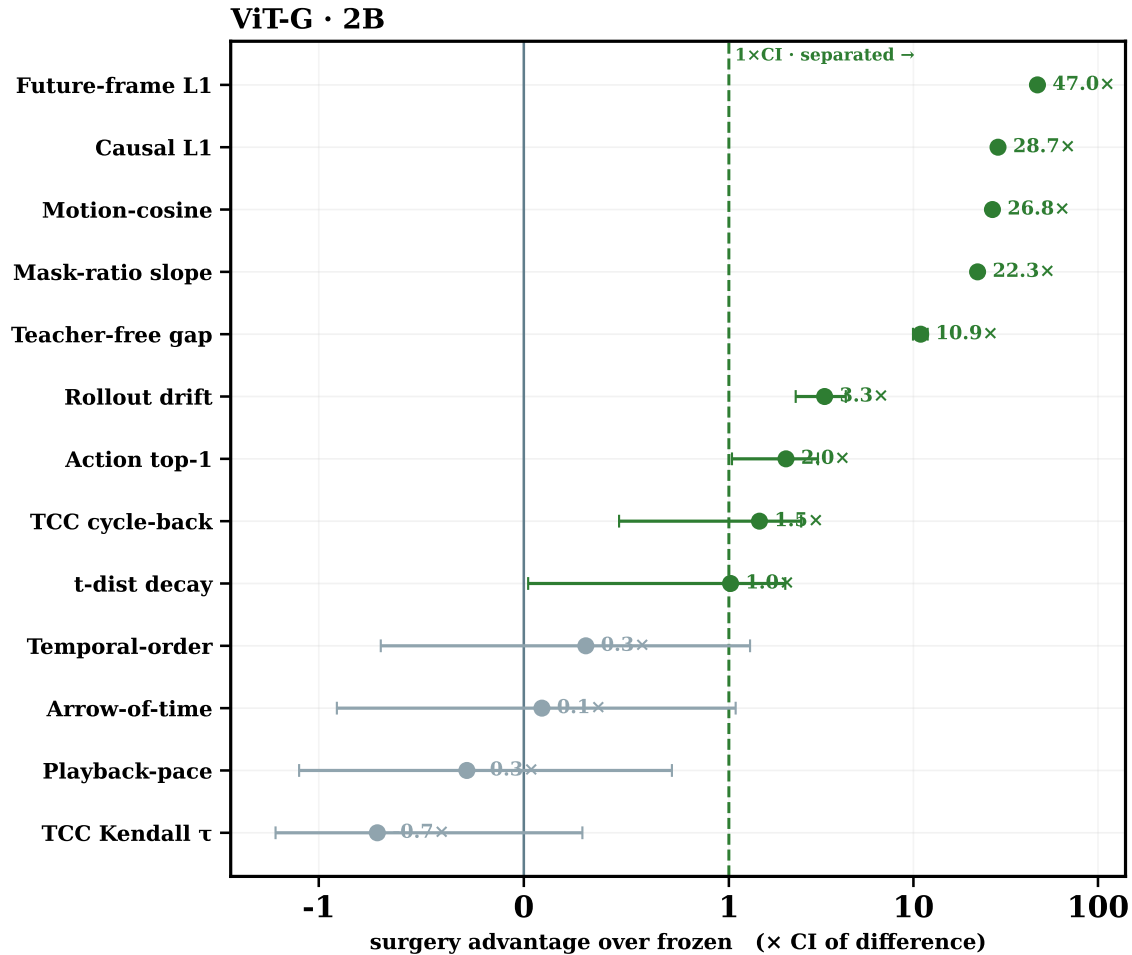
Forest plot — does surgery's MEAN beat the best competitor? (green = yes · surgery_raw ∈ OURs)



- Magnitude view: the predictive lead is small-but-consistent -- future-L1 +1.7%/+1.4%, mask-ratio +1.1%/+2.9%, teacher-free +3.8% (1B).
- The motion-cosine deficit is large (-44.9% / -33.4%) -- a deliberate trade: surgery optimizes prediction, full-FT optimizes semantics.

3 · THE PAPER CLAIM - surgery vs FROZEN (CI-widths)

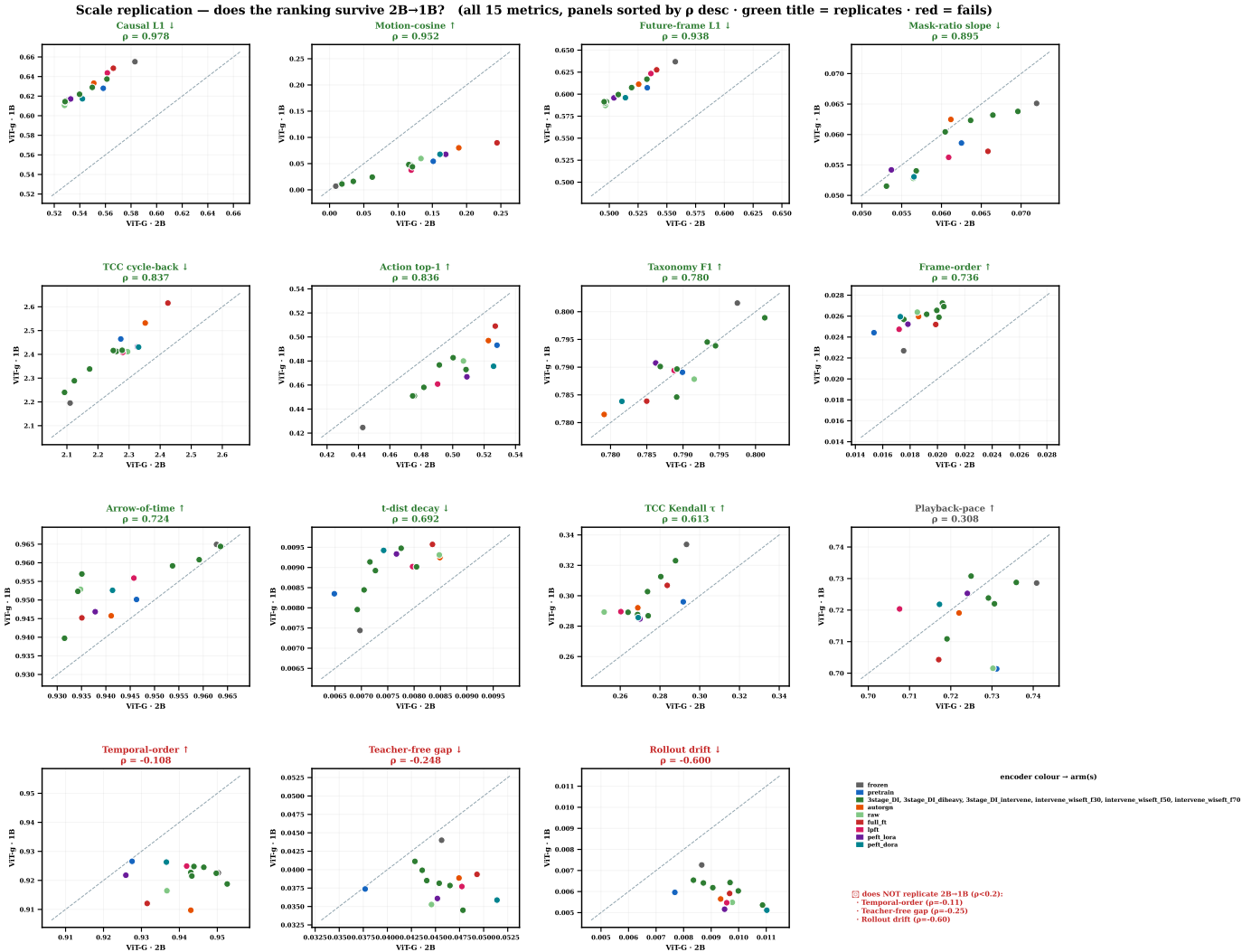
Forest plot — does surgery statistically separate from frozen? (green past dashed line = yes · surgery_raw ∈ OURs)



● Decisive: surgery separates from frozen by 5-47 CI-widths on EVERY predictive + motion metric (future-L1 47x/27x, motion-cos 27x/26x) at BOTH scales.

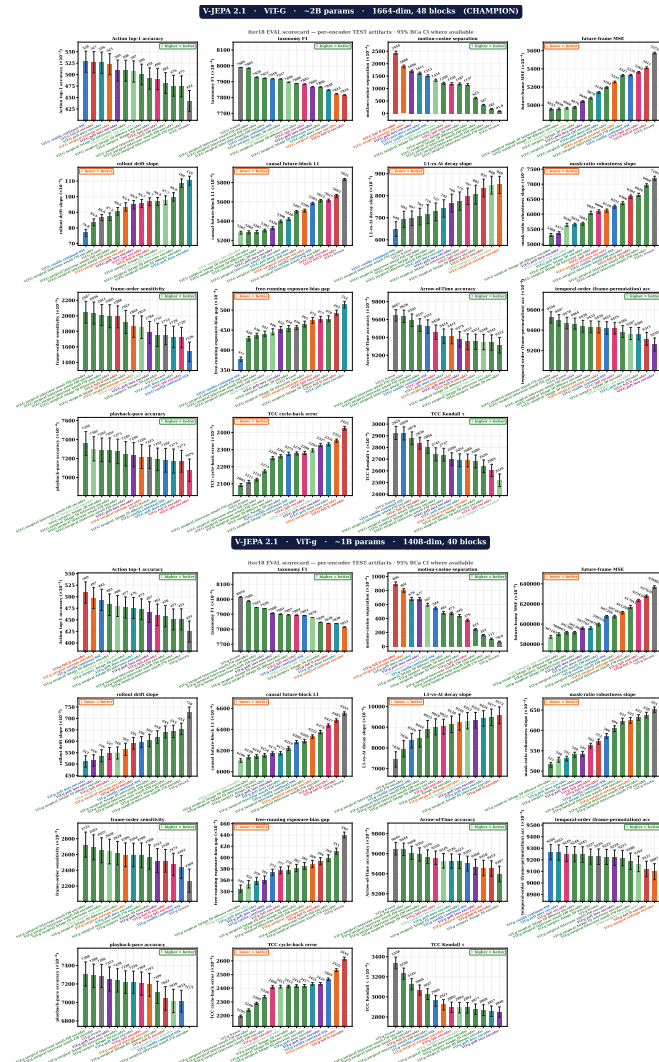
● The cost is coherence -- surgery regresses on TCC-tau / TCC-cycle (grey/negative), the frame-timing metrics frozen still owns; a stated trade.

4 · Scale transfer - does the 1B rank like the 2B? (Spearman rho)



- Core metrics replicate 2B->1B: causal-L1 rho=0.978, motion-cos 0.952, future-L1 0.938, mask-ratio 0.895 -> the cheap 1B is a faithful proxy for the 2B.
- 12/15 metrics replicate (rho>0.2); 3 secondary ones fail (rollout -0.60, teacher-free -0.25, temporal-order -0.11) -- flagged, not over-claimed.

5 · Full appendix scorecard - 2B champion + 1B, all 15 metrics



- Every arm x 15 metrics at both scales (n=1,825, 95% BCa CI): the same three-behaviour split -- surgery wins prediction, full-FT semantics, frozen coherence.
- The 1B (bottom) is a scaled-down twin of the 2B champion (top) -- the bar orderings mirror, which is exactly what scale_replication quantifies.